

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-016: Show / Hide Password Visibility Toggle in Login and Add Student Modules

Field	Details
Change Request ID	CR-016
Change Request Title	Show / Hide Password Visibility Toggle in Login and Add Student Modules
Student Name	N. Bavanika
Student ID	JFICT24_19
Implementation Status	Completed
Date	8 October 2026

1. Change Request Summary

1.1 Business Requirement

Users and administrators currently enter passwords into masked input fields without an option to view or verify the entered text. As passwords often contain numbers, uppercase and lowercase letters, and special characters, users may make typing errors without realizing it.

The system should provide an intuitive password visibility toggle that allows users to show or hide the entered password before submission. The same functionality should also be available in the Administrator's Add Student form to improve password entry accuracy while preserving the existing validation, authentication, and student creation functionality.

1.2 Existing Behaviour

Before implementation:

- LoginPage.jsx renders LoginFormSection.jsx with username and password.
- The login password input was statically configured with type="password".
- Password characters were always masked.
- No password visibility toggle button or eye icon was available in the Login form.
- The existing authentication sends credentials through POST /api/auth/login.
- The Administrator Add Student form also displayed the password as a masked field.
- No password visibility toggle was available in the Add Student password field.
- Existing password validation and student creation functionality were already available.

1.3 Requested Change

Add a client-side show/hide password toggle to the Login form and Administrator Add Student form.

The implementation must:

- Toggle the Login password input between "password" and "text".
- Toggle the Add Student password input between "password" and "text".
- Keep the password masked by default when the forms are loaded.

- Display an intuitive eye icon when the password is hidden.
- Display an eye-slash icon when the password is visible.
- Include accessible aria-label and aria-pressed attributes.
- Ensure the toggle button uses type="button" so it does not submit the form.
- Provide sufficient right-side input padding so the password text does not overlap the icon.
- Support keyboard interaction through the button.
- Preserve existing Login validation and error handling.
- Preserve existing Add Student password validation.
- Pass the original password value correctly regardless of visibility state.
- Maintain the existing Administrator and Student authentication flow.
- Maintain the existing Add Student API and student creation functionality.
- Avoid changes to the backend authentication, password hashing, database structure, or API contracts.

2. Implementation

2.1 Implementation Summary

CR-016 was implemented as a frontend usability enhancement for password input fields.

A password visibility state was added to the Login form and Administrator Add Student form. The password input type changes between "password" and "text" when the user activates the visibility toggle.

The implementation uses React state to control password visibility. The password value itself is not modified when the visibility state changes.

An eye icon is displayed while the password is hidden, and an eye-slash icon is displayed while the password is visible.

The implementation also includes accessibility attributes and prevents accidental form submission by using type="button" for the visibility toggle.

The existing authentication, password validation, student creation, API requests, and role-based navigation were preserved.

2.2 Affected Files

File	Change	Purpose
frontend/src/components/LoginFormSection.jsx	Modified	Added password visibility state, show/hide toggle button, eye icons, and accessibility attributes.
frontend/src/components/StudentFormModal.jsx	Modified	Added the same password visibility toggle to the Administrator Add Student password field.
frontend/src/styles/login.css	Modified	Added styling for password wrapper, toggle button, icon positioning,

		focus state, and input spacing.
frontend/src/pages/LoginPage.jsx	No functional change	Existing login validation, authentication request, and role-based navigation remain unchanged.
Backend files	No change	No backend authentication or API logic was modified.
Database	No change	No database schema or data changes were required.

2.3 Technical Changes

Password Visibility State

A React state variable was added to control the password visibility:

```
const [showPassword, setShowPassword] = useState(false);
```

The default value is false, ensuring that the password remains masked when the form is initially loaded.

Dynamic Password Input Type

The password input type is dynamically controlled:

```
type={showPassword ? "text" : "password"}
```

When showPassword is false, the password remains masked.

When showPassword is true, the entered password becomes visible.

Toggle Button

A dedicated button was added beside the password input.

The button uses:

```
type="button"
```

This prevents the visibility control from triggering form submission.

The button toggles the visibility state without changing the actual password value.

Accessibility

The visibility button includes:

```
aria-label={
  showPassword
```

```
    ? "Hide password"  
    : "Show password"  
}
```

and:

```
aria-pressed={showPassword}
```

These attributes provide an accessible description of the current password visibility state.

Icon Behaviour

Two visual states are provided:

- Eye icon – password is currently hidden.
- Eye-slash icon – password is currently visible.

The icon changes automatically according to the showPassword state.

Input Layout

The password input was given additional right-side padding to prevent entered text from overlapping the visibility button.

The password toggle button is positioned inside the password input area using the .password-input-wrapper.

Browser Password Reveal Icon

Browser-provided password reveal controls were also suppressed where applicable to avoid displaying two visibility icons at the same time.

The application therefore provides one consistent password visibility control.

2.4 Implementation Logic

1. Load the Login or Add Student form.
2. Initialize showPassword as false.
3. Display the password input using type="password".
4. Display the eye icon.
5. User enters the password normally.
6. User activates the eye button.
7. Update showPassword to true.
8. Change the input type to "text".
9. Display the eye-slash icon.
10. The original password value remains unchanged.
11. User activates the button again.
12. Change the input type back to "password".
13. Display the eye icon again.
14. Submit the form normally using the existing form submission logic.

15. Existing validation, authentication, and student creation behaviour continues unchanged.

2.5 Error and Data Handling

Existing error handling was preserved.

For Login:

- Empty username/password validation remains unchanged.
- Existing login error messages remain unchanged.
- Authentication continues using the existing login(username, password) function.
- Administrator users continue to navigate to /dashboard.
- Student users continue to navigate to /student-dashboard.

For Add Student:

- Existing password validation remains unchanged.
- Minimum password length and password complexity validation remain unchanged.
- Existing Add Student API request remains unchanged.
- The password value submitted to the backend is not altered by changing its visibility.
- Existing student creation and error handling remain unchanged.

No password is stored in React state in a new format or transmitted through a new API.

3. Testing & Verification

3.1 Testing Approach

The CR-016 implementation was verified through manual frontend testing.

Testing Types:

- Functional testing
- Positive testing
- Validation testing
- UI testing
- Accessibility testing
- Keyboard interaction testing
- Regression testing
- Login authentication testing
- Add Student functionality testing
- Responsive UI testing
- Error handling testing

Environment: Development

Browser: Chrome

Test Role: Administrator

Test Date: 8 October 2026

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open Login page and inspect password field	Password is masked by default	Password was masked by default	Pass
TC-02	Enter password in Login form	Password characters remain hidden	Password remained masked	Pass
TC-03	Click Login eye icon	Password becomes visible	Password became visible	Pass
TC-04	Click eye-slash icon	Password becomes hidden again	Password was hidden again	Pass
TC-05	Verify icon state	Eye/eye-slash changes according to visibility	Icon changed correctly	Pass
TC-06	Toggle password without submitting Login form	Toggle should not submit the form	Form was not submitted	Pass
TC-07	Verify accessibility attributes	aria-label and aria-pressed are correctly provided	Attributes worked correctly	Pass
TC-08	Use keyboard to focus and activate toggle	Toggle can be activated using keyboard	Toggle worked using keyboard	Pass
TC-09	Enter valid credentials while password is masked	Login should succeed	Login succeeded	Pass
TC-10	Enter valid credentials while password is visible	Login should succeed with same password	Login succeeded	Pass
TC-11	Enter invalid credentials	Existing login error should be displayed	Error message displayed correctly	Pass
TC-12	Open Add Student form	Password is masked by default	Password was masked	Pass
TC-13	Enter password in Add Student form	Password remains masked	Password remained masked	Pass
TC-14	Click Add Student eye icon	Password becomes visible	Password became visible	Pass
TC-15	Click eye-slash icon in Add Student form	Password becomes masked again	Password was masked again	Pass
TC-16	Submit Add Student form with visible password	Student should be created using the entered password	Student creation worked correctly	Pass
TC-17	Submit Add Student form with invalid/empty password	Existing validation should be displayed	Existing validation worked correctly	Pass
TC-18	Verify password text does not overlap icon	Password text and icon should remain visually separated	No overlap observed	Pass
TC-19	Verify existing Login role navigation	Admin and Student redirects should remain unchanged	Existing redirects worked correctly	Pass
TC-20	Verify responsive Login UI	Password field and toggle should remain usable on smaller screens	Responsive layout worked correctly	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	20
Passed	20
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

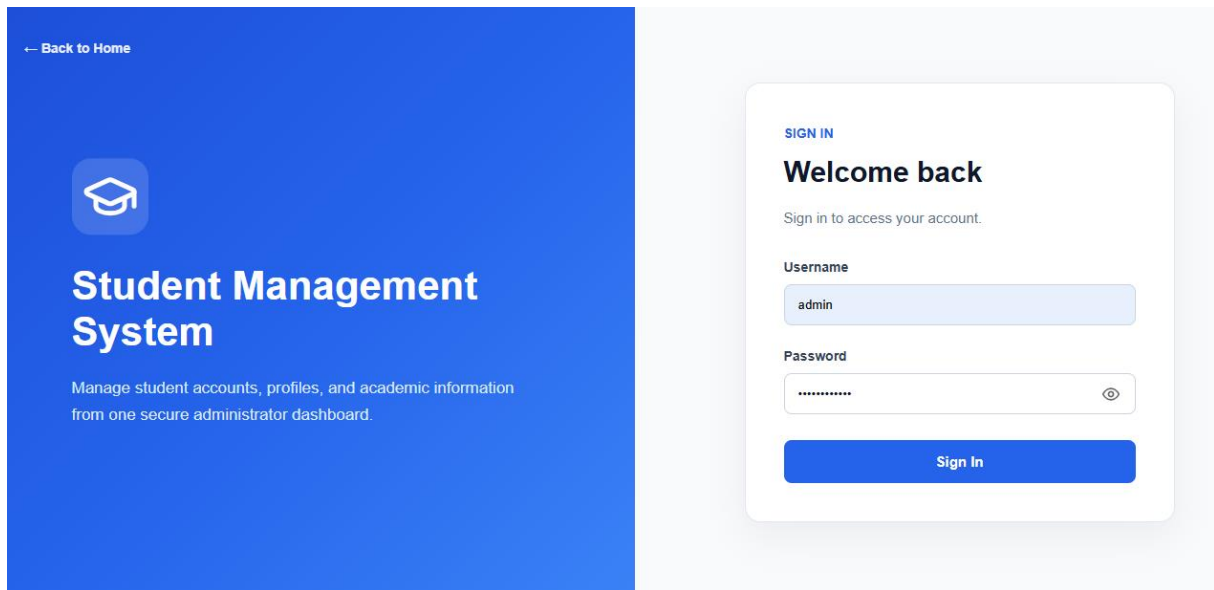
ID	Acceptance Criterion	Verification	Status
AC-001	Login password is masked by default.	Login initial-state test	Met
AC-002	User can show and hide the Login password using the toggle.	Login toggle functional test	Met
AC-003	Eye and eye-slash icons correctly indicate password visibility.	UI verification	Met
AC-004	Toggle does not submit the Login form.	Form submission verification	Met
AC-005	Toggle provides appropriate accessibility attributes.	aria-label / aria-pressed verification	Met
AC-006	Password text does not overlap the visibility icon.	UI/layout verification	Met
AC-007	Valid Login credentials work when password is masked or visible.	Authentication testing	Met
AC-008	Existing Login errors and Administrator/Student navigation continue to work.	Regression testing	Met
AC-009	Administrator Add Student password field provides a show/hide toggle.	Add Student functional test	Met
AC-010	Add Student password value remains correct regardless of visibility state.	Add Student submission test	Met
AC-011	Existing Add Student password validation remains unchanged.	Validation testing	Met
AC-012	Existing Add Student functionality continues to work after the enhancement.	Add Student regression test	Met

5. Evidence

Evidence 01 – Login Password Visibility Toggle

Related: AC-001, AC-002, AC-003, AC-004, AC-006, AC-007

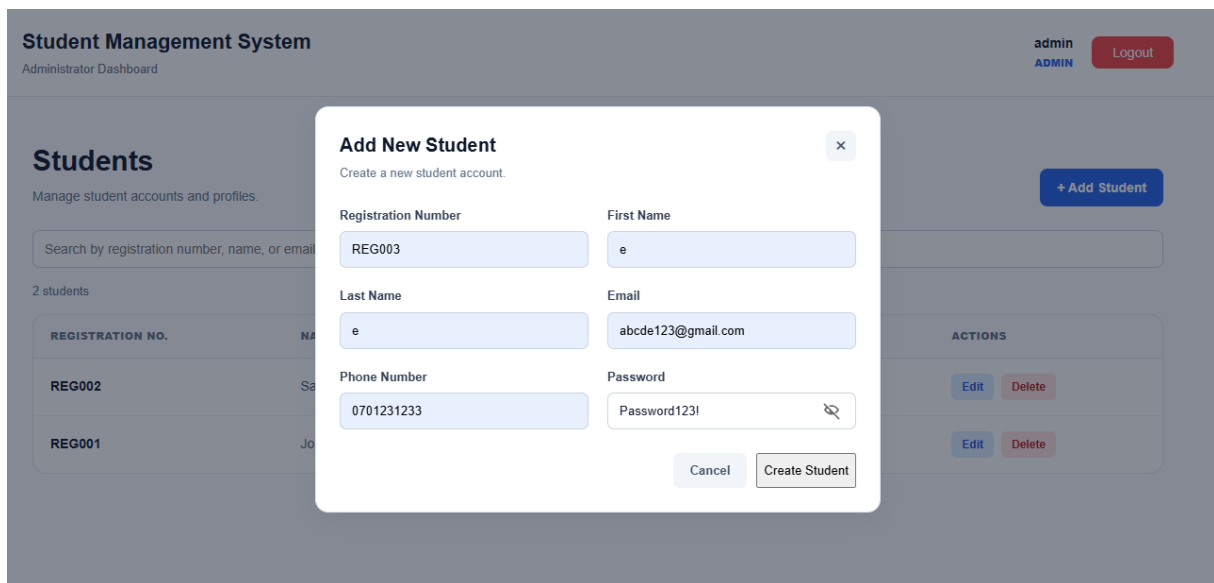
Figure 1: Login page showing the password field with the password visibility toggle. The screenshot should demonstrate the visible password state and eye icon.



Evidence 02 – Administrator Add Student Password Toggle

Related: AC-009, AC-010, AC-011, AC-012

Figure 2: Administrator Add Student form showing the password visibility toggle and the visible password state.



6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Existing Login Authentication	Verified
Existing Add Student Functionality	Verified
Backend/API Changes	None Required
Database Changes	None Required

Final Status	Accepted
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6.1 Final Summary

CR-016 was successfully implemented as a frontend password visibility enhancement for the Login and Administrator Add Student modules.

The implementation allows users to show and hide entered passwords using an eye/eye-slash toggle while keeping the password masked by default. Accessibility attributes, keyboard interaction, input spacing, and form submission behaviour were addressed.

The same functionality was added to the Administrator Add Student password field without changing existing password validation, student creation logic, API requests, authentication, or database functionality.

A total of 20 functional, validation, UI, accessibility, regression, and authentication test cases were executed. All 20 test cases passed, resulting in a 100% pass rate.

All defined acceptance criteria were verified and marked as met.

6.2 Issues / Limitations

No significant issues identified.

The password visibility functionality is a presentation-level frontend enhancement. When the user intentionally activates the visibility toggle, the password is displayed in plaintext on screen. The password remains masked by default, and no new password storage, logging, authentication, or database mechanism was introduced.

Student Name: N.Bavanika

Student ID: JFICT24_19

Date: 8 October 2026

Signature:

~ END OF REPORT ~